

The 1PN Correction to the Chirp-Mass Frequency Evolution

Part II: Extending the Quadrupole-Order Derivation

1 Recap and New Assumptions

Part I (`chirp_mass_derivation.tex`) derived, at leading (“Newtonian” / quadrupole / 0PN) order,

$$\dot{f}_{GW} = \frac{96}{5}\pi^{8/3} \left(\frac{GM}{c^3} \right)^{5/3} f_{GW}^{11/3}, \quad \mathcal{M} \equiv \mu^{3/5} M^{2/5}, \quad (1)$$

using (i) the Newtonian circular orbit, (ii) Kepler’s third law $\omega^2 = GM/a^3$, and (iii) the leading-order quadrupole radiation formula. Here we extend this to the next order in the post-Newtonian (PN) expansion: relativistic corrections of relative order $v^2/c^2 \sim GM/(ac^2)$ to *both* the orbital dynamics and the radiation.

New/updated assumptions:

1. Same binary configuration as Part I: point masses m_1, m_2 , circular orbit, adiabatic inspiral. We keep the same definitions $M = m_1 + m_2$, $\mu = m_1 m_2 / M$, $\eta = \mu / M$ (symmetric mass ratio, $0 < \eta \leq 1/4$), $\mathcal{M} = \mu^{3/5} M^{2/5}$.
2. We introduce the standard dimensionless **PN expansion parameter**

$$x \equiv \left(\frac{GM\omega_{\text{orb}}}{c^3} \right)^{2/3}, \quad (2)$$

which reduces to $x = v^2/c^2$ for the relative orbital velocity $v = \omega_{\text{orb}} a$ at leading (Newtonian/Keplerian) order, so $x \ll 1$ is the formal PN bookkeeping parameter. One can check x reduces to the Newtonian variable used implicitly in Part I: $Mc^2 x = M(GM\omega)^{2/3}$, and $\frac{1}{2}M\eta c^2 x = \frac{1}{2}\mu(GM)^{2/3}\omega^{2/3} = -E_{\text{orb}}^{(N)}$ exactly reproduces Eq. (6) of Part I.

3. **Harmonic gauge**, standard in PN computations of the two-body problem (as opposed to the TT gauge used for the outgoing radiation itself).
4. We *do not* re-derive the 1PN equations of motion or the 1PN radiative multipole moments from the Einstein field equations here — that computation (post-Newtonian expansion of the harmonic-gauge field equations, matching of near-zone and wave-zone metrics, the Einstein–Infeld–Hoffmann equations of motion, and the 1PN mass quadrupole plus leading current-quadrupole/mass-octupole flux contributions) is a substantial calculation in its own right and is beyond hand-derivation scope for a school. Exactly as we took the *leading-order* quadrupole formula as a given input in Part I, we here take the well-established **1PN-accurate energy and flux** as inputs (Wagoner & Will 1976; corrected by Blanchet & Schäfer 1989; see also Blanchet, *Living Rev. Relativity* for a modern derivation and full reference chain). What we *will* derive in full, step by step, is how these known $E(x)$, $\mathcal{F}(x)$ combine — via the same energy-balance argument as Part I — into the 1PN-corrected chirp formula.

2 Input: 1PN-Accurate Orbital Energy and Flux

For a circular orbit, the standard PN results (see references above) are

$$E(x) = -\frac{1}{2}M\eta c^2 x \left[1 - \frac{9+\eta}{12}x + \mathcal{O}(x^2) \right], \quad (3)$$

$$\mathcal{F}(x) = \frac{32c^5}{5G}\eta^2 x^5 \left[1 - \left(\frac{1247}{336} + \frac{35}{12}\eta \right) x + \mathcal{O}(x^2) \right]. \quad (4)$$

Sanity check against Part I: setting the bracketed PN-correction terms to zero (i.e. truncating at x^0) must reproduce the Newtonian results. Indeed $-\frac{1}{2}M\eta c^2 x = -\frac{1}{2}\mu(GM)^{2/3}\omega^{2/3}$ matches $E_{\text{orb}}(\omega)$ from Part I exactly, and using $\eta^2 M^{10/3} = \mu^2 M^{4/3} = \mathcal{M}^{10/3}$ (Part I, Eq. 9) one finds $\mathcal{F}(x) \rightarrow \frac{32}{5}(G^{7/3}/c^5)\mathcal{M}^{10/3}\omega^{10/3}$, exactly Eq. (10) of Part I. This cross-check confirms the normalization/convention of x is consistent with our Part I variables.

3 Energy Balance to 1PN

As in Part I, adiabatic radiation reaction enforces

$$\frac{dE}{dt} = \frac{dE}{dx} \frac{dx}{dt} = -\mathcal{F}(x). \quad (5)$$

3.1 Step 1: Differentiate $E(x)$

Expand Eq. (3):

$$E(x) = -\frac{1}{2}M\eta c^2 x + \frac{1}{24}M\eta c^2(9+\eta)x^2 + \mathcal{O}(x^3). \quad (6)$$

Differentiating term by term,

$$\frac{dE}{dx} = -\frac{1}{2}M\eta c^2 + \frac{1}{12}M\eta c^2(9+\eta)x + \mathcal{O}(x^2) = -\frac{1}{2}M\eta c^2 \left[1 - \frac{9+\eta}{6}x \right] + \mathcal{O}(x^2). \quad (7)$$

(Note the coefficient doubles from 1/12 to 1/6 upon factoring out $-\frac{1}{2}$, since $\frac{1}{12}M\eta c^2(9+\eta) = -\frac{1}{2}M\eta c^2 \cdot \left(-\frac{9+\eta}{6}\right)$.)

3.2 Step 2: Solve for $\dot{x}$

From Eq. (5),

$$\dot{x} = \frac{-\mathcal{F}(x)}{dE/dx} = \frac{\mathcal{F}(x)}{\frac{1}{2}M\eta c^2 \left[1 - \frac{9+\eta}{6}x \right]}. \quad (8)$$

Insert Eq. (4):

$$\dot{x} = \frac{2}{M\eta c^2} \cdot \frac{32c^5}{5G}\eta^2 x^5 \left[1 - \left(\frac{1247}{336} + \frac{35}{12}\eta \right) x \right] \left[1 - \frac{9+\eta}{6}x \right]^{-1}. \quad (9)$$

3.3 Step 3: Expand consistently to $\mathcal{O}(x)$ beyond leading order

Since we only trust $E(x)$ and $\mathcal{F}(x)$ to relative order x (1PN), we must expand the quotient consistently and *drop* any term of relative order x^2 : using $[1 - \epsilon x]^{-1} = 1 + \epsilon x + \mathcal{O}(x^2)$ with $\epsilon = (9 + \eta)/6$,

$$\begin{aligned}\dot{x} &= \frac{64c^3\eta}{5GM} x^5 \left[1 - \left(\frac{1247}{336} + \frac{35}{12}\eta \right) x \right] \left[1 + \frac{9+\eta}{6}x \right] + \mathcal{O}(x^7) \\ &= \frac{64c^3\eta}{5GM} x^5 \left[1 + \left(\frac{9+\eta}{6} - \frac{1247}{336} - \frac{35}{12}\eta \right) x \right] + \mathcal{O}(x^7).\end{aligned}\tag{10}$$

Combine over the common denominator 336 (using $\frac{9+\eta}{6} = \frac{504+56\eta}{336}$ and $\frac{35}{12}\eta = \frac{980\eta}{336}$):

$$\frac{504+56\eta}{336} - \frac{1247}{336} - \frac{980\eta}{336} = \frac{-743-924\eta}{336} = -\left(\frac{743}{336} + \frac{11}{4}\eta \right),\tag{11}$$

where we used $924/336 = 11/4$. Hence

$$\boxed{\dot{x} = \frac{64c^3\eta}{5GM} x^5 \left[1 - \left(\frac{743}{336} + \frac{11}{4}\eta \right) x \right]}.\tag{12}$$

4 From $\dot{x}$ to $\dot{f}_{GW}$

4.1 Chain rule: $x \rightarrow \omega$

From the definition Eq. (2), $x = (GM/c^3)^{2/3}\omega^{2/3}$, so

$$\frac{dx}{d\omega} = \frac{2}{3} \left(\frac{GM}{c^3} \right)^{2/3} \omega^{-1/3} = \frac{2x}{3\omega} \implies \dot{\omega} = \frac{3\omega}{2x} \dot{x}.\tag{13}$$

Substituting Eq. (12):

$$\dot{\omega} = \frac{3\omega}{2x} \cdot \frac{64c^3\eta}{5GM} x^5 \left[1 - \left(\frac{743}{336} + \frac{11}{4}\eta \right) x \right] = \frac{96c^3\eta}{5GM} \omega x^4 \left[1 - \left(\frac{743}{336} + \frac{11}{4}\eta \right) x \right].\tag{14}$$

Using $x^4 = (GM/c^3)^{8/3}\omega^{8/3}$, so $\omega x^4 = (GM/c^3)^{8/3}\omega^{11/3}$:

$$\dot{\omega} = \frac{96c^3\eta}{5GM} (GM/c^3)^{8/3} \omega^{11/3} \left[1 - \left(\frac{743}{336} + \frac{11}{4}\eta \right) x \right] = \frac{96}{5} \eta \frac{(GM)^{5/3}}{c^5} \omega^{11/3} \left[1 - \left(\frac{743}{336} + \frac{11}{4}\eta \right) x \right].\tag{15}$$

4.2 Collapse to the chirp mass

Exactly as in Part I ($\mathcal{M}^{5/3} = \mu M^{2/3} = \eta M^{5/3}$), the Newtonian-order prefactor collapses to $\mathcal{M}$:

$$\eta (GM)^{5/3} = G^{5/3} \eta M^{5/3} = G^{5/3} \mathcal{M}^{5/3},\tag{16}$$

so

$$\dot{\omega}_{\text{orb}} = \frac{96}{5} \frac{G^{5/3}}{c^5} \mathcal{M}^{5/3} \omega_{\text{orb}}^{11/3} \left[1 - \left(\frac{743}{336} + \frac{11}{4}\eta \right) x \right].\tag{17}$$

4.3 Convert to GW frequency

With $\omega_{\text{orb}} = \pi f_{GW}$ as in Part I:

$$\boxed{\dot{f}_{GW} = \frac{96}{5} \pi^{8/3} \left(\frac{GM}{c^3} \right)^{5/3} f_{GW}^{11/3} \left[1 - \left(\frac{743}{336} + \frac{11}{4} \eta \right) x \right], \quad x = \left(\frac{\pi GM f_{GW}}{c^3} \right)^{2/3}.} \quad (18)$$

This is the **1PN chirp equation**. It reduces to the Part I Newtonian result exactly when the bracket is set to 1 (i.e. dropping the $\mathcal{O}(x)$ term). This is the leading term of the widely used *TaylorF2* PN frequency-domain phasing formula.

Physical remark: a qualitatively new feature

At Newtonian (0PN) order, $\dot{f}_{GW}$ depended on the two component masses *only* through the single combination $\mathcal{M}$ — a strict degeneracy between m_1 and m_2 at fixed $\mathcal{M}$. At 1PN order, the correction term depends on $\eta = \mu/M$ *independently* of $\mathcal{M}$. This is the first PN order at which the frequency evolution starts to break the chirp-mass degeneracy and becomes sensitive to the mass *ratio* — physically, this is why higher-order PN (and full IMR) templates are needed to measure m_1 and m_2 separately, rather than just $\mathcal{M}$.

5 Time to Coalescence at 1PN

Following the same integration strategy as Part I §6(a), separate Eq. (12) and expand the integrand consistently to $\mathcal{O}(x)$:

$$dt = \frac{5GM}{64c^3\eta} x^{-5} \left[1 - \left(\frac{743}{336} + \frac{11}{4} \eta \right) x \right]^{-1} dx \approx \frac{5GM}{64c^3\eta} \left\{ x^{-5} + \left(\frac{743}{336} + \frac{11}{4} \eta \right) x^{-4} \right\} dx. \quad (19)$$

Integrating from x (time t) to the formal coalescence point $x \rightarrow \infty$ (time t_c), term by term ($\int_x^\infty x'^{-5} dx' = \frac{1}{4} x^{-4}$, $\int_x^\infty x'^{-4} dx' = \frac{1}{3} x^{-3}$):

$$\tau(x) \equiv t_c - t = \frac{5GM}{64c^3\eta} \left[\frac{1}{4} x^{-4} + \frac{1}{3} \left(\frac{743}{336} + \frac{11}{4} \eta \right) x^{-3} \right] = \frac{5GM}{256c^3\eta} x^{-4} \left[1 + \frac{4}{3} \left(\frac{743}{336} + \frac{11}{4} \eta \right) x \right]. \quad (20)$$

Using the same chirp-mass identity as before to eliminate $(M, \eta) \rightarrow (\mathcal{M}, \eta)$ in the leading prefactor ($GM/\eta \cdot x^{-4} = c^8 G^{-5/3} \mathcal{M}^{-5/3} (\pi f)^{-8/3}$, exactly as derived in Part I),

$$\boxed{\tau(f) = \frac{5}{256} \left(\frac{GM}{c^3} \right)^{-5/3} (\pi f)^{-8/3} \left[1 + \frac{4}{3} \left(\frac{743}{336} + \frac{11}{4} \eta \right) x \right], \quad x = \left(\frac{\pi GM f}{c^3} \right)^{2/3}.} \quad (21)$$

Setting the bracket to 1 reproduces Part I, Eq. (21), as required.

6 Summary

$$E(x) = -\frac{1}{2} M \eta c^2 x \left[1 - \frac{9+\eta}{12} x \right], \quad \mathcal{F}(x) = \frac{32c^5}{5G} \eta^2 x^5 \left[1 - \left(\frac{1247}{336} + \frac{35}{12} \eta \right) x \right], \quad (22)$$

$$\dot{f}_{GW} = \frac{96}{5} \pi^{8/3} \left(\frac{GM}{c^3} \right)^{5/3} f_{GW}^{11/3} \left[1 - \left(\frac{743}{336} + \frac{11}{4} \eta \right) x \right], \quad \tau(f) = \frac{5}{256} \left(\frac{GM}{c^3} \right)^{-5/3} (\pi f)^{-8/3} \left[1 + \frac{4}{3} \left(\frac{743}{336} + \frac{11}{4} \eta \right) x \right] \quad (23)$$

with $x = (\pi GMf/c^3)^{2/3}$, $\eta = \mu/M$. Both reduce to the Part I Newtonian (quadrupole-only) results when $x \rightarrow 0$.

Validity: the PN expansion is an expansion in $x \sim v^2/c^2$; it is only asymptotic and becomes unreliable as the binary approaches merger ($x \sim 1$, i.e. $a \sim$ a few GM/c^2), which is why numerical relativity / full IMR waveform models are needed for the late inspiral–merger–ringdown, while the PN expansion is excellent in the early inspiral where $x \ll 1$.